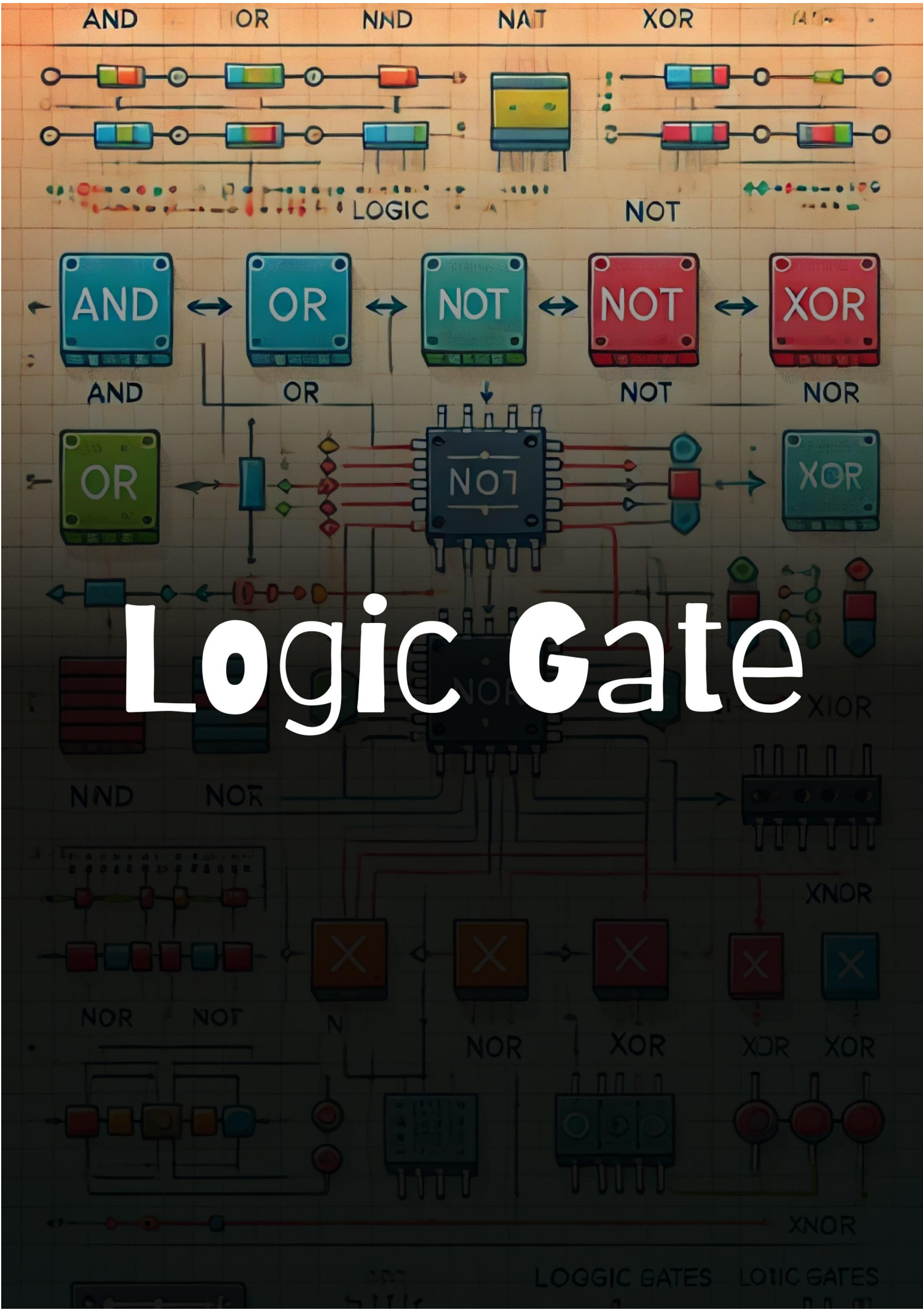

AND
OR
NAND
NAT
XOR
LOGIC
NOT
AND
OR
NOT
NOT
XOR
AND
OR
NOT
NOR
OR
NOT
XOR
NAND
NOR
XNOR
NOR
NOT
NOR
XOR
XOR
XOR
XNOR
LOGGIC GATES
LOGIC GATES

Logic Gate

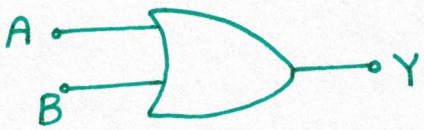
logic Gate

A digital circuit which either stops a signal or allows to pass through it is called a gate.
Logic gate has 1 or more input but only one output.
There are three basic logic gates:

1.) OR Gate
2.) AND Gate
3.) NOT Gate

OR GATE

0: low
1: high



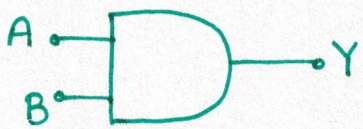
A	B	Y=A+B
0	0	0
1	0	1
0	1	1
1	1	1

Boolean Algebra

0+0 = 0
1+0 = 1
0+1 = 1
1+1 = 1

(Higher one will sent forward

AND GATE



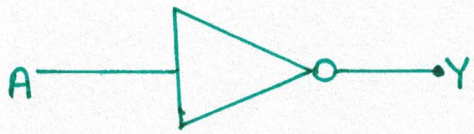
Both must be present to get output

A	B	Y=A.B
0	0	0
1	0	0
0	1	0
1	1	1

Boolean Algebra

0.0 = 0
1.0 = 0
0.1 = 0
1.1 = 0

NOT GATE



A	$Y = \overline{A}$
0	1
1	0

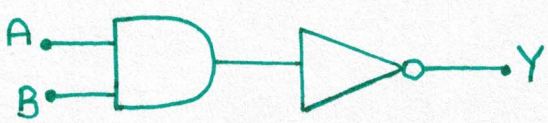
Boolean Algebra

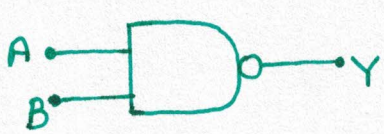
$\overline{0} = 1$

$\overline{1} = 0$

COMBINATION OF GATES

NAND GATE (NOT + AND)





A	B	$Y = \overline{A.B}$
0	0	1
1	0	1
0	1	1
1	1	0

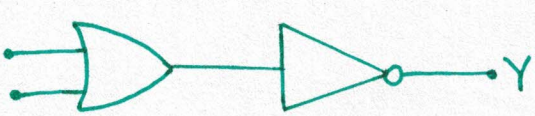
$\overline{0.0} = 1$

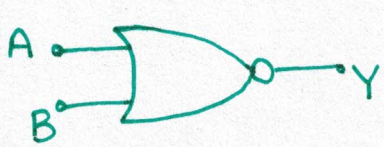
$\overline{1.0} = 1$

$\overline{0.1} = 1$

$\overline{1.1} = 0$

NOR GATE (NOT + OR)





A	B	$Y = \overline{A+B}$
0	0	1
1	0	0
0	1	0
1	1	0

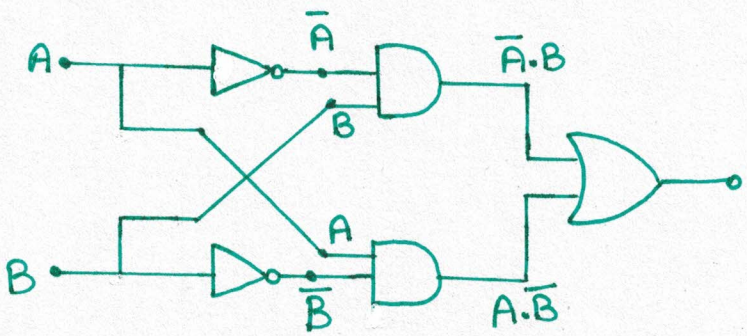
$\overline{0+0} = 1$

$\overline{1+0} = 0$

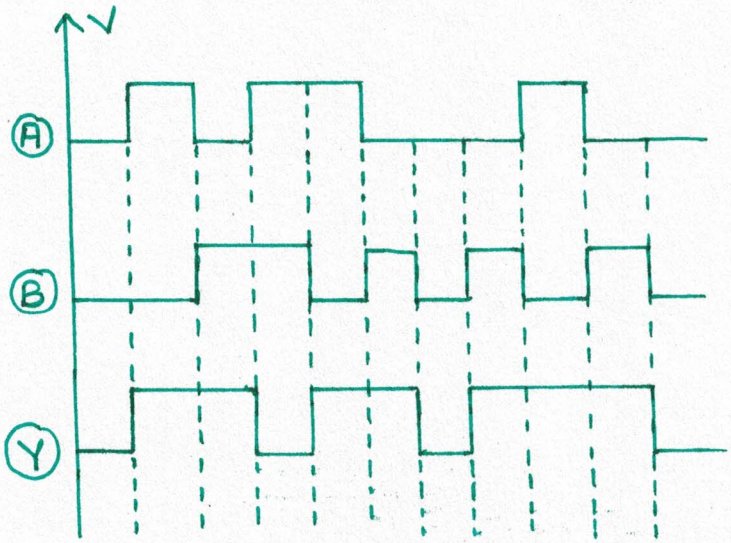
$\overline{0+1} = 0$

$\overline{1+1} = 0$

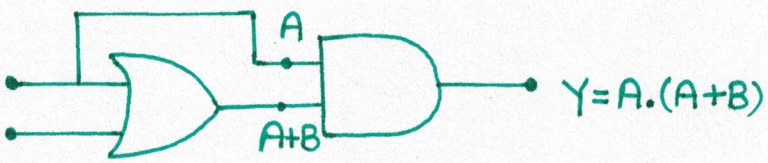
XOR GATE



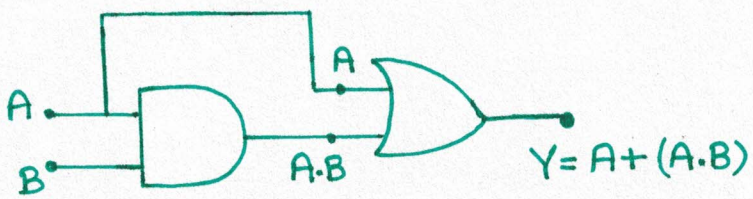
$Y = \overline{A}B + A\overline{B}$



A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0



A	B	Y
0	0	0
1	0	1
0	1	0
1	1	1



A	B	Y
0	0	0
1	0	1
0	1	0
1	1	1